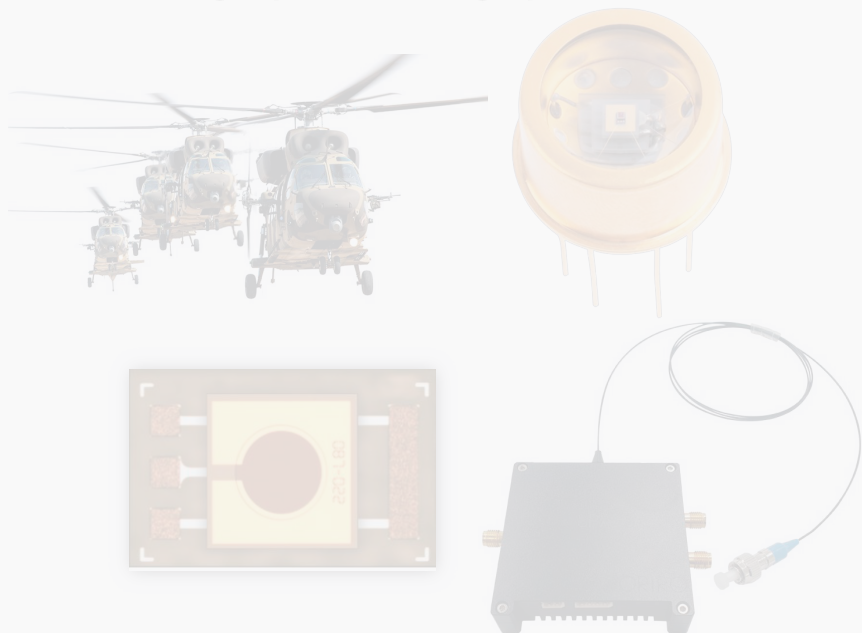


SWIR Sensor Technologies for Defense and Security

- From single photon to high power detection -



WOORIRO Co. Ltd.

(Electro-optic Component Specialists)

ABOUT WOORIRO

Wooriro was founded in 1998 and began producing Planar Lightwave Circuit (PLC) devices and PLC related devices such as optical splitters and Arrayed Waveguide Grating (AWG) device. Since the introduction Wooriro PLC products to the optical communication market, we believe that we've contributed significantly to the FTTH networks worldwide. Inevitably, we became the world-wide leader in PLC splitter.

Wooriro expanded its production of InGaAs Photodiode (PD) for electro-optical communication both in commercial and defense starting 2010. It would later develop series of electro-optical modules such as, 10Gb/s PIN-PD/APD, 25Gb/s PIN-PD/APD, 50Gb/s PIN-PD/APD chip and ROSA. Wooriro continuously refined InGaAs APD technology, and as a result have successfully developed InGaAs Single Photon Avalanche Diode (SPAD). It proved to provide the best performance in today's quantum cryptography communication hardware market. Wooriro is primed to enhance the performance of defense sensors requirement greatly. We are confident that our single-photon technology will bring advancement to the defense industry.

Wooriro will continue to improve the InGaAs sensor technology and production of electro-optical modules for advanced security and defense systems. Consequently, Wooriro strives to provide superior products to the future defense and security systems.

Dr. Chan-Yong Park, Vice President and CTO

cypark@wooriro.com

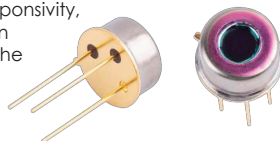
Large Diameter InGaAs PIN PD for Laser Monitoring

(Model : WLAPD3K/WLAPD2K/WLAPD1K)

InGaAs PIN PD

Description

WLAPD3K(2K) is a photodetector which has a large light-receiving surface diameter of 3mm(2mm). Since it is InGaAs/InP compound semiconductor, it can detect 1000 ~ 1650nm wavelength laser signal, and convert optical signal into electrical signal. The OE-conversion efficiency, called responsivity, is higher than 1A/W at 1550nm and 0.8A/W at 1064nm. It also has an excellent reliability with high shunt resistance resulting in low noise. The WLAPD3K is used for laser signal monitoring.



Applications

- Laser signal monitoring
- Optical pulse detection for Laser warning system
- SWIR sensor
- Optical power meter

Features

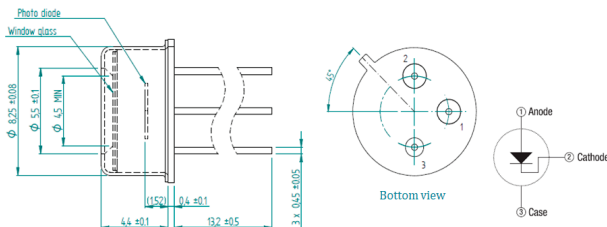
- 800nm ~ 1700nm spectral response
- High Sensitivity
- High Shunt Resistance
- Very Low Dark Current
- Low noise and high reliability
- Standard TO-5 package with flat window

Optical and Electrical Specifications (WLAPD3K)¹⁾

Parameter	Symbol	Test Conditions	Min.	Typ.	Max.	Unit
Light-receiving Diameter ¹⁾	D	-		3		mm
Wavelength Range	λ		800		1700	nm
Shunt Resistance	Rsh		10	30		M Ω
Dark Current (per element)	I _d	VR=Vop		20	200	nA
Responsivity at 1064nm	R		0.7	0.8		A/W
Responsivity at 1550nm	R		0.85	1.05		A/W
Operating voltage	Vop			2	5	V
Linearity	-	VR=2V(@1300nm), -45~5dBm		±5		%
Capacitance	Cd	@VR=Vop		600		pF
Operating Temperature	To		-40		85	°C
package type				TO-5		

¹⁾ light-receiving diameters of WLAPD2K&WLAPD1K are 2mm&1mm. The performances are not shown here.

Dimension



InGaAs Quadrant PD (Model : WQPD3K/WQPD2K)

InGaAs PIN PD

► Description

QPD, a large surface area InGaAs photodiode was developed to improve tracking capability in Semi-Active Laser(SAL) missile seeker guidance technology. QPD's surface has four high sensitivity light receiving area which minimizes cross-talk among the receiving areas. QPD also has wide spectral response range in the area of short wavelength infrared and possesses highly effective signal detection rate compared to Silicon based photodiode. QPDs are used in position sensor, beam alignment, beam profiling, signal tracking & stabilization in mobile space communication, and increased usage in automatic position control system for autonomous vehicles.



► Applications

- Laser seeker
- Position Sensing
- Laser spot tracking
- Laser range finder
- Free space optical communication
- Missile Guidance
- Beam Alignment
- Optical Tweezers
- Auto tracking system

► Features

- 800nm ~ 1700nm spectral response
- High Sensitivity
- Low Crosstalk
- High Shunt Resistance
- Low Dark Current
- Low noise and high reliability
- Monolithic chip with minimal dead space
- Standard TO-5 package with flat window

► Optical and Electrical Specifications (WQPD3K)¹⁾

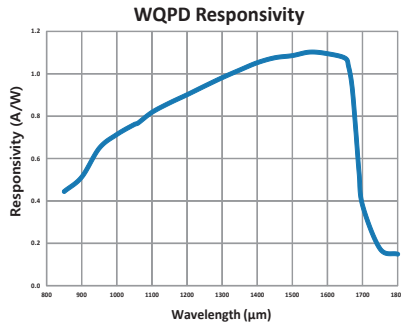
Parameter	Symbol	Test Conditions	Min.	Typ.	Max.	Unit
Light-receiving Diameter ¹⁾	D	-		3		mm
Gap between Elements	d			0.025		mm
Crosstalk		$V_R=V_{op}$		<1%	2%	
Wavelength Range	λ		800		1700	nm
Shunt Resistance	Rsh			30		MΩ
Dark Current (per element)	I _d	$V_R=V_{op}$		2	70	nA
Responsivity at 1064nm	R		0.7	0.75		A/W
Responsivity at 1550nm	R		0.9	1.05		A/W
Operating voltage	V _{op}			2	5	V
Noise equivalent power	NEP	@ 1300 nm		1		pW/√Hz
Capacitance	C _d	per quadrant, at $V_R=V_{op}$		100		pF
Operating Temperature	T _o		-40		85	°C
package type				TO-5		

¹⁾ light-receiving diameter of WQPD2K is 2mm . The performance data is available upon request.

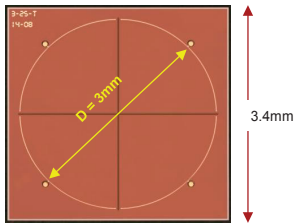
InGaAs Quadrant PD (Model : WQPD3K/WQPD2K)

InGaAs PIN PD

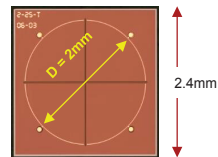
► Spectral Responsivity



► Chip Dimension

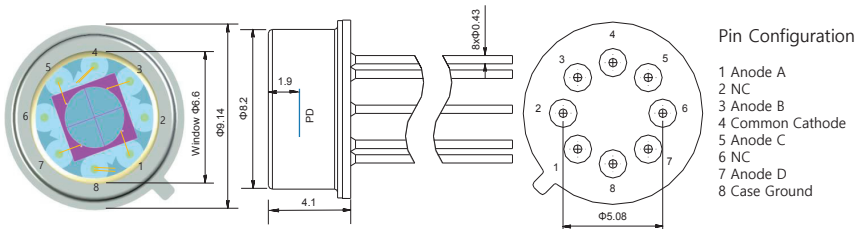


<WQPD3K>



<WQPD2K>

► Package Dimension



※ Note : The specifications can be modified per customer's request.

InGaAs APD Chip (Model : WLAAPD200/WLAAPD80)

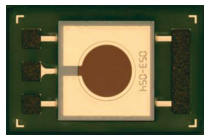
InGaAs Avalanche Photodiode

Description

InGaAs avalanche photodiodes (APD) with diameters of 200 μ m and 80 μ m are produced for highly sensitive LRF receivers to detect in the SWIR range. Since APD has an internal current gain (multiplication), it can be more sensitively operated than PIN, normally over 10 times. There are 4 types of APD chips. WLAAPD200F and WLAAPD80F are front-illumination APDs. WLAAPD200B and WLAAPD80B are bottom-illumination APDs. The bottom-illumination APD employs round-trip of the light signal and responsivity is better than front-illumination by 20-25%. Furthermore, the capacitance is smaller than the front due to absence of p-bonding pad, the operation speed and sensitivity are better than the front-illumination. The bottom-illumination APD chips are flip-chip bonded to quartz substrate. The corresponding TO cans are also available.



WLAAPD200F



WLAAPD200B

Product Code	200 μ m Diameter	80 μ m Diameter
Front illumination Chip	WLAAPD200F	WLAAPD80F
Bottom illumination CoC	WLAAPD200B	WLAAPD80B

Applications

- Eye-safe Laser Rangefinder (LRF) receiver
- Eye-safe LiDAR/LADAR receiver
- Optical sensor for SWIR detection
- Optical Time-domain Reflectometer (OTDR)
- Free space optical communication

Features

- 1000nm ~ 1700nm spectral response
- High Sensitivity
- Low Dark Current
- Low noise and high reliability
- Excellent endurance for high optical power
- Very small capacitance and high speed
- Standard TO46 package with flat window

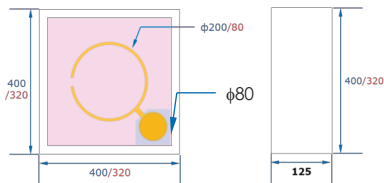
Optical and Electrical Specifications (Typ. @25 °C)

Parameter	Symbol	Test Conditions	Min.	Typ.	Max.	Unit
Breakdown voltage	V_{BR}	$I_D = 100 \mu A$, $T = 25^\circ C$		60		V
Dark current	I_D	$V_R = 0.9 V_{BR}$		5		nA
Capacitance (200F)	C_{APD}	$f = 1 \text{ MHz}$, $V = 0.9 V_{br}$		2.4		pF
Capacitance (200B)				1.7		
Capacitance (80F)				0.65		
Capacitance (80B)				0.35		
Responsivity	R	$V_R = 0.97 V_{BR}$, $\lambda = 1550 \text{ nm}$		10		A/W
Temperature coefficient	γ	$\Delta V_{br} / \Delta T$		0.1		V/°C
Operating Temperature	T_o		-40		85	°C
package type				TO-5		

InGaAs APD chip (Model : WLAAPD200/WLAAPD80)

InGaAs Avalanche Photodiode

► Chip and CoC(Chip-on-Carrier) dimension (unit : μm)

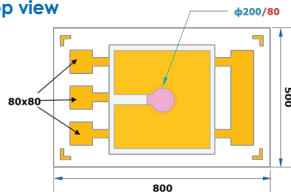


Top view

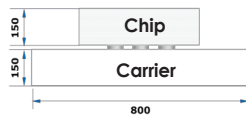
Side view

Front-illumination Chip

Top view

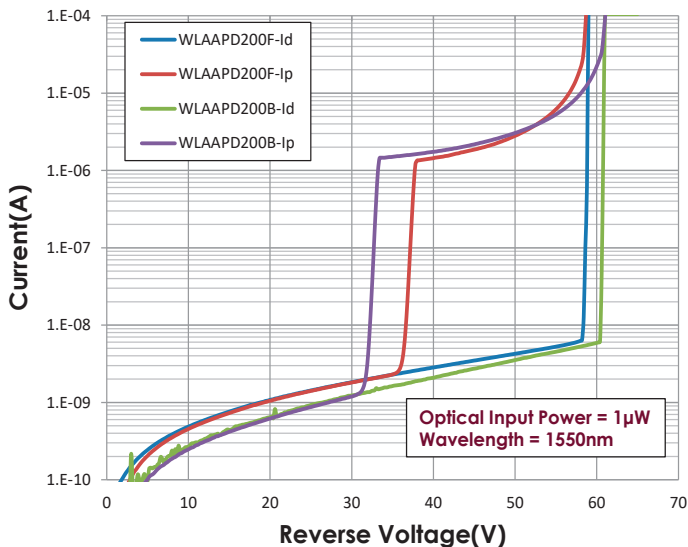


Side view



Bottom-illumination CoC

► Current-Voltage properties



Laser Rangefinder Receiver (Model : WDR110)

InGaAs APD Receiver

► Description

Model WDR110 eye-safe LRF receiver is produced for Laser Rangefinder receiver and area surveying using photon. The InGaAs APD is highly sensitive to photons and provide accuracy of the projected images as well as safe environment for the users. CMOS IC is designed for more complex performance and compactness. Single FR4 PCB and standard TO-8 package provides a low-cost solution for LRF receiver.



► Applications

- Laser rangefinder of Eye-safe wavelength range
- Eye-safe LiDAR / LADAR

► Features

- 5ns ~ 28ns input pulse
- 200um InGaAs APD chip
- Integrated CMOS TIA and LOGIC
- Providing TPT(Time Programmed Threshold) mode
- Internally adjusted and temperature compensated APD bias
- Standard TO-8, FR4 PCB and quartz window for low-cost application
- Low cost, Low power consumption

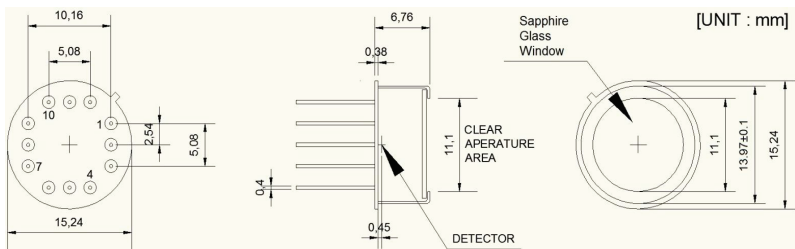
► Specifications

Parameter		Performance	Condition
PD	Type	InGaAs APD	
	Diameter/Cap	200um/2.4pF	
	Leakage current	Compensated	
MDS(minimum detectable signal)		10 nW type 15 nW max	@5ns pulse, MR=5A/W, SNR=3.3 (PD=50%, FAR=0.1%) Noise=4.0mVrms max
Multiple target resolution		200ns max	@100uW and 5ns pulse, successive pulses with same amplitude
Time programmed threshold		Enable	External circuit needed
Dynamic Range		1:55,000	From noise level to 100uW
Walk error		10ns typ / 20ns max	@5ns pulse
Power consumption		3.3V / 17mA	

Laser Rangefinder Receiver (Model : WDR110)

InGaAs APD Receiver

► Package dimension (unit : mm)



No.	Symbol	Description
1	VTH1	Control False Alarm Rate.
2	VTH2	Control False Alarm Rate.
3	VTH CTRL	Control TPG(Time Programmed Gain)
4	VSS	GND
5	APD3.3V	Power Supply Voltage for the APD Bias circuits.
6	DETECT	Outputs a detection signal.
7	VDDA	Power Supply Voltage for the Analog circuits.
8	+HV	High Voltage VDD of APD Bias. +75V typ.
9	VAPD TP	VAPD Bias TP.
10	INT OUT TP	Integrator output TP for Optical Alignment.
11	VDDD	Power Supply Voltage for the Digital circuits.
12	VTH1 HALF	Control False Alarm Rate.

► Absolute Maximum Ratings

Parameter	Symbol	Rating	Unit
TIA supply voltage	V_{DD}	+3.6	V
APD supply voltage	V_{HV}	75	V
Optical input power	P_{opt}	1	mW
Operating case temperature range	T_C	-32 to +64	°C
Storage temperature range	T_{STG}	-40 to +85	°C

InGaAs SPAD for Quantum Cryptography

(Model : WSPAD)

InGaAs Geiger-mode APD

Description

WSPAD is Single Photon Avalanche Diode made of InGaAs compound semiconductor to detect 1550nm single photon range. Since the SPAD chip can multiply single photon carrier by millions of times, it can detect a signal composed of single photon. Generally, it is used as a detector for quantum cryptography, especially in Quantum Key Distribution (QKD) network. WSPAD shows excellent performance in very low dark count rate (DCR) noise, high detection efficiency, good after-pulse probability (APP). There are two package types. Mini-flat type (WSPAD-M) has advantages of surface-mountable and better heat-dissipation capable compared to TO-8 (WSPAD-C).



Mini-flat type
(WSPAD-M)



TO-8 Coaxial
type (WSPAD-C)

Applications

- Quantum Cryptography (QKD)
- Single Photon Counting
- Optical Time Domain Reflectometry
- Long Range LRF(Laser Rangefinder)
- Fundamental Studies in Quantum Physics

Features

- 1000nm ~ 1650nm spectral response
- Operation up to GHz Gating frequency
- Built-in 3-stage TEC(Thermo-Electric Cooler)
- Very Low Dark Count Rate (< kHz)
- High Single Photon Detection Efficiency

Specifications

	Parameter	Symbol	Test Conditions	Min.	Typ.	M ax.	Unit
Linear Mode (+25℃)	Breakdown Voltage	V_B	$I_D = 100 \mu A$	50	70	90	V
	Capacitance	C	$F = 1 \text{ MHz}, V_{PB} = 0.9V_B$	-	0.25	-	pF
	Temperature Dependence of V_B	γ	Between -40℃ and 85℃, linear approximation	-	0.11	-	V/℃
Geiger Mode (-40℃)	AfterPulse Probability	APP	10MHz Gate Frequency 2ns Gate Pulse 20% PDE	-	-	10	%
	Dark Count Rate	DCR		-	0.5(P)	-	KHz
	Detection Efficiency	PDE		-	20	-	%

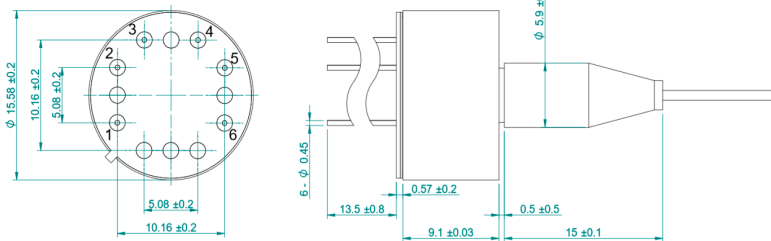
InGaAs SPAD for Quantum Cryptography

(Model : WSPAD)

InGaAs Geiger-mode APD

► Package dimension (unit : mm)

Note: Mini-flat packages are not shown here.
please make a separate inquiry



No.	Symbol	Description (Bottom View)
1	TEC	TEC(-)
2	TEC	TEC(+)
3	A	APD Anode
4	C	APD Cathode
5	TH	Thermistor1
6	TH	Thermistor2

• Fiber length : 1 m (standard) , Connector : FC-PC (Standard)

► ABSOLUTE MAXIMUM RATING

Parameter	Conditions	Max	Unit
Forward Current	Continuous bias	+ 1	mA
Forward Voltage	Continuous bias	+ 1	V
Reverse Current	Continuous bias	- 1	mA
Reverse Voltage	Continuous bias	- V_B	V
	Pulsed bias (gated mode)	- ($V_B + 10$)	V
Optical Input Power	Continuous wave	1	mW
TEC Current		2.4	A
TEC Voltage		1.9	V
Thermistor	@25°C	2.2 / 10.0	kΩ

InGaAs NFAD for Quantum Sensing (Model : WNFAAD)

InGaAs Geiger-mode APD

► Description

WNFAAD is Negative Feedback Avalanche Diode consisting of a SPAD and a resistor. It is made by adding quenching resistor in the external circuit of the SPAD. Since the NFAD chip can multiply single photon carrier by hundred thousand times or better, it can detect a signal composed of single photon to several photons. It also detects irregular incident photon signals. NFAD is used as detector for quantum sensing such as single photon counting, long-distance Light Detection and Ranging (LiDAR) and Laser Rangefinder (LRF). There are two package types. Mini-flat type(WNFAAD-M) has advantages of surface-mountable and better heat-dissipation capability compared to TO-8(WNFAAD-C).



Mini-flat type
(WNFAAD-M)



TO-8 Coaxial
type (WNFAAD-C)

► Applications

- Quantum Cryptography (QKD)
- Single Photon Counting
- Optical Time Domain Reflectometry
- Long Range LRF(Laser Rangefinder)
- Fundamental Studies in Quantum Physics

► Features

- 1000nm ~ 1650nm spectral response
- Suitable to free-running operation
- Built-in Quenching resistor (Optional value)
- Built-in 3-stage TEC(Thermo-Electric Cooler)
- TO can / Fiber-pigtail (Multi Mode Fiber)

► Specifications

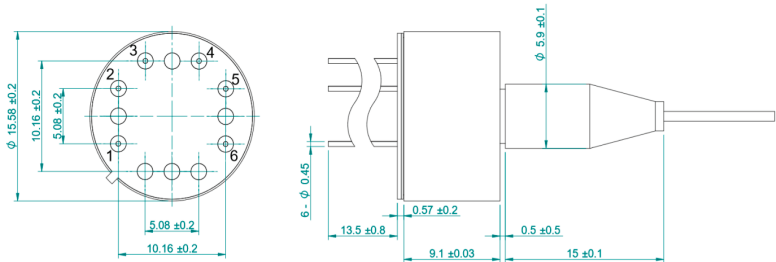
	Parameter	Symbol	Test Conditions	Min.	Typ.	Max.	Unit
Linear Mode (+25°C)	Breakdown Voltage	V_B	$I_D = 100 \text{ mA}$	50	70	90	V
	Capacitance	C	$F = 1 \text{ MHz}$, $V_{PB} = 0.9V_B$	-	0.25	-	pF
	Temperature Dependence of V_B	g	Between -40°C and 85°C, linear approximation	-	0.11	-	V/K
Geiger mode (-40°C)	Timing Jitter	TJ		-	-	500	ps
	Dark Count Rate	DCR	PDE= 10%	-	50	-	KHz
	Detection Efficiency	PDE	$V_{ex} = 2.0V$	10	-	-	%

InGaAs NFAD for Quantum Sensing (Model : WNFA0)

InGaAs Geiger-mode APD

► Package dimension (unit : mm)

Remark : Mini-flat package not shown here.
For mini-flat, please inquire separately



No.	Symbol	Description (Bottom View)
1	TEC	TEC(-)
2	TEC	TEC(+)
3	A	APD Anode
4	C	APD Cathode
5	TH	Thermistor1
6	TH	Thermistor2

• Fiber length : 1m (standard) , Connector : FC-PC (Standard) / Flat window TO-can type (Optional)

► ABSOLUTE MAXIMUM RATING

Parameter	Conditions	Max	Unit
Forward Current	Continuous bias	+ 1	mA
Forward Voltage	Continuous bias	+ 1	V
Reverse Current	Continuous bias	- 1	mA
Reverse Voltage	Continuous bias	- V_B	V
	Excess bias	- ($V_B + 5$)	V
Optical Input Power	Continuous wave	1	mW
TEC Current		2.4	A
TEC Voltage		1.9	V
Thermistor	@25°C	2.2 / 10.0	kΩ

InGaAs Solid-State Photomultiplier (Model : WSSPM)

InGaAs Geiger-mode APD

Description

Solid State Photomultiplier (SSPM) is an InGaAs based device that counts photon numbers in excruciating detail. It is constructed with hundreds of NFAD chip array as pixel of SSPM. Because it consists of hundreds of InGaAs /InP SPAD pixels with corresponding negative feedback resistors and integrated micro-lens, SSPM can detect single photon and multi-photons, also analyze or count photon numbers. While all the pixels with the corresponding negative feedback resistors are connected to the common anode and cathode, every pixel boost its signal power several tens of thousands times independently during single or multiple photons incident on the multiple pixels at the same time. The each current generated in multiple pixels would add up, which means that SSPM can distinct or count the number of incident photons. This is very useful properties for SSPM be applicable to LiDAR receiver or ultra-sensitive receiver sensor such as gas-sensors and bio-sensors.



Applications

- Single Photon LiDAR of SWIR range
- Photon Number Resolver
- Security and Safety monitoring
- Long Range LRF (Laser Rangefinder)
- Gas Sensor
- Bio-Medical Imaging

Features

- High FoV with integrate InP micro-lens array
- Hundreds of NFAD Pixels
- Operates at Eye-safe SWIR
- High detection efficiency due to round-trip of incident photon
- Built-in 3-stage TEC (Thermo-Electric Cooler)

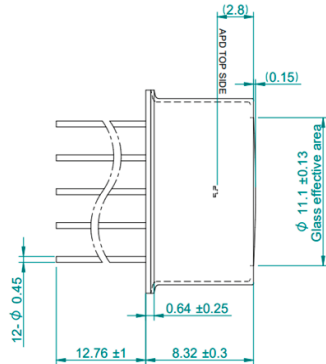
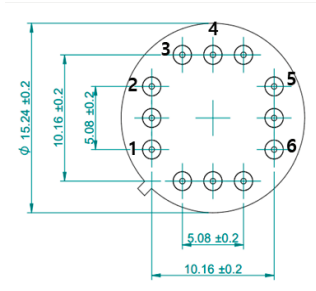
Specifications

	Parameter	Symbol	Test Conditions	Min.	Typ.	Max.	Unit
Linear Mode (+25°C)	Breakdown Voltage	V_B	$I_D = 100 \mu A$	50	70	90	V
	Capacitance of one pixel	C	$F = 1 \text{ MHz}, V_{PD} = 0.9V_B$	-	0.25	-	pF
	Temperature Dependence of V_B	γ	Between -40°C and 85°C, linear approximation	-	0.11	-	V/K
Geiger mode (-20°C)	Dead Time	Td		1.5	-	-	ns
	Dark Count Rate	DCR	PDE= 10%	-	25	-	MHz
	Detection Efficiency	PDE	$V_{ex} = 2.0V$	15	-	-	%

InGaAs Solid-State Photomultiplier (Model : WSSPM)

InGaAs Geiger-mode APD

► Package dimension (unit : mm)



No.	Symbol	Description (Bottom View)
1	TH	Thermistor1
2	TH	Thermistor2
3	A	APD Anode
4	C	APD Cathode
5	TEC	TEC(-)
6	TEC	TEC(+)

► ABSOLUTE MAXIMUM RATING

Parameter	Conditions	Max	Unit
Forward Current	Continuous bias	+ 1	mA
Forward Voltage	Continuous bias	+ 1	V
Reverse Current	Continuous bias	- 1	mA
Reverse Voltage	Continuous bias	- V_B	V
	Excess bias	- ($V_B + 5$)	V
Optical Input Power	Continuous wave	1	mW
TEC Current		2.4	A
TEC Voltage		1.9	V
Thermistor	@25°C	2.2	kΩ

2.5G APD Receiver for Free Space Optical Communication (Model : WFS2.5GA)

Receiver for FSO

Description

WFS2.5GA is a receiver for optical communication use. It can be supplied as an optical receiver with 105 μ m multimode fiber(MMF)-pigtailed module or TO-can for an easy use for receiver of free space optical communication. It combines an InGaAs avalanche photodiode(APD) and Trans-impedance amplifier (TIA) that operates at the data rate of 2.5Gb/s. The APD-TIA TO-CAN or MMF-pigtailed receiver shows very high sensitivity and extremely reliable properties.



105/125 μ m
MMF-
pigtailed type



TO46 can
type

Applications

- Receiver for free space optical communication
- High-speed fiber optic communication system

Features

- InGaAs APD / 1000nm ~ 1650nm
- High Sensitivity of -39dBm @ 2.488Gb/s and 10^{-6} BER [when ER of light source equals to 10dB]
- 105 μ m MMF pigtail for a large aperture
- Combination of aspherical lens plus integrated Indium Phosphide micro lens for a large aperture.

ABSOLUTE MAXIMUM RATING

Parameter	Symbol	Rating	Unit
TIA supply voltage	V_{CC}	-0.4 to 4	V
APD supply voltage	V_{APD}	0 to V_{BR}	V
APD reverse current	I_{APD}	2	mA
Operating case / Storage temperature range	T_C/T_{STG}	-40 to +85	$^{\circ}$ C

Optical and Electrical Specifications (T=25 $^{\circ}$ C)

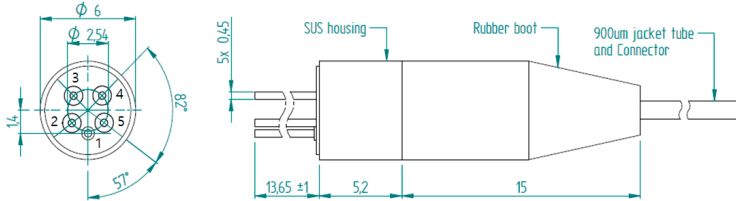
Parameters	Symbol	Test Condition	Min.	Typ.	Max.	Unit
Dark current	I_d	$0.9V_{BR}$		1	20	nA
Optical Wavelength Range	λ	-	1000		1650	nm
Sensitivity	P_s	2.488 Gb/s, PRBS=2 ²³ -1, ER=10dB, λ =1550nm, BER=10 ⁻⁶		-39	-37	dBm
Overload	P_{MAX}		-4			dBm
TIA Supply Voltage	V_{CC}	-	2.97	3.3	3.46	V
Breakdown Voltage	V_{BR}	Dark current, I_d =10mA	40	60	80	V
Temperature Coefficient of V_{BR}	γ	Operating case temperature		0.1		V/ $^{\circ}$ C

2.5G APD Receiver for Free Space Optical Communication (Model : WFS2.5GA)

Receiver for FSO

► Package dimension (unit : mm)

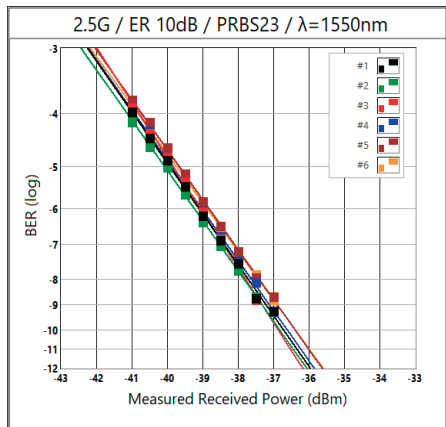
Remark : MMF-pigtailed only shown.



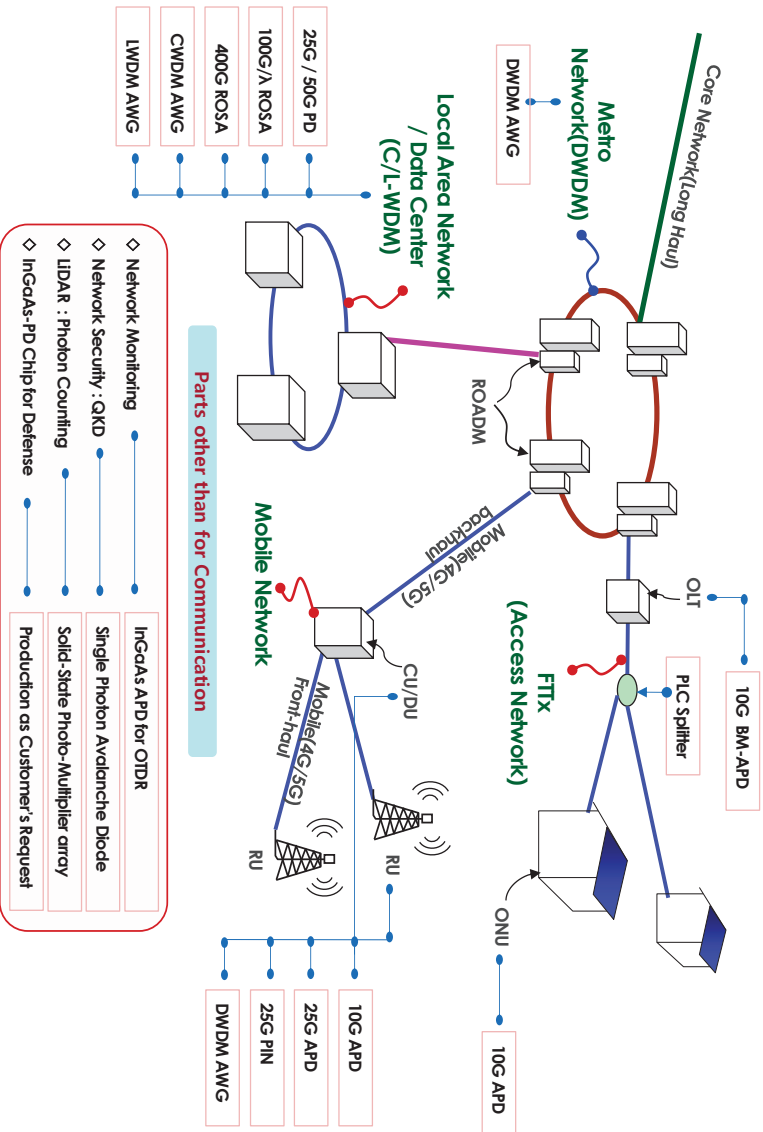
No.	Pin name	Type	Description
1	GND	Ground	Ground
2	DATA	RF+ output	TIA positive output
3	V _{CC}	Power	TIA supply voltage
4	V _{APD}	Power	APD supply voltage
5	$\overline{\text{DATA}}$	RF- output	TIA negative output

- Fiber length : 1m (standard) , Connector type : SC-PC (standard)
- Fiber length and connector type are available as customizing

► BER(Bit Error Rate) curve



WOORIRO Products for Network and Sensor Application



Check List

DATE . .

[illegible]



wooriro

WOORIRO Co., Ltd.



Wooriro Co., Ltd. (H.Q)
102-22, Pyeongdongsandan 6beon-ro
Gwangsan-gu, Gwangju, 62453
Republic of Korea

Wooriro USA
1952 Gallows Road, Suite 110
Vienna, VA 22182
U.S.A



(H.Q) support@wooriro.com/ +82-62-602-8202
(USA) pkemp@wooriro.com/ +1-571-474-9006



www.wooriro.com